In[74]:=
```
A = {{0, 1}, {1, 0}};
B = {{1, 0}, {1, 1}};
{MatrixForm [A], MatrixForm [B]}
```

Out[76]=
$$\left\{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}\right\}$$

In[77]:=
```
I2 = IdentityMatrix [2];
```

In[78]:=
```
Q = (A - x I2) . (A - x I2) + ConjugateTranspose [(B - (y + i z) I2)] . (B - (y + i z) I2) ;
Q = FullSimplify [Q, x ∈ Reals && y ∈ Reals && z ∈ Reals];
```

In[80]:=
```
MatrixForm [Q]
```

Out[80]//MatrixForm=
$$\begin{pmatrix} 3 + x^2 + (-2 + y)\, y + z^2 & 1 - 2\, x - y - i\, z \\ 1 - 2\, x - y + i\, z & 2 + x^2 + (-2 + y)\, y + z^2 \end{pmatrix}$$

In[81]:=
```
{evalQ , evecQ} = Eigensystem [Q]
```

Out[81]=
$$\left\{\left\{\frac{1}{2} \times \left(5 + 2\, x^2 - 4\, y + 2\, y^2 + 2\, z^2 - \sqrt{5 - 16\, x + 16\, x^2 - 8\, y + 16\, x\, y + 4\, y^2 + 4\, z^2}\right),\right.\right.$$
$$\left.\frac{1}{2} \times \left(5 + 2\, x^2 - 4\, y + 2\, y^2 + 2\, z^2 + \sqrt{5 - 16\, x + 16\, x^2 - 8\, y + 16\, x\, y + 4\, y^2 + 4\, z^2}\right)\right\},$$
$$\left\{\left\{-\frac{1 - \sqrt{5 - 16\, x + 16\, x^2 - 8\, y + 16\, x\, y + 4\, y^2 + 4\, z^2}}{2 \times (-1 + 2\, x + y - i\, z)},\ 1\right\},\right.$$
$$\left.\left.\left\{-\frac{1 + \sqrt{5 - 16\, x + 16\, x^2 - 8\, y + 16\, x\, y + 4\, y^2 + 4\, z^2}}{2 \times (-1 + 2\, x + y - i\, z)},\ 1\right\}\right\}\right\}$$

In[82]:=
```
sMin = Simplify [evalQ[[1]], x ∈ Reals && y ∈ Reals && z ∈ Reals]
```

Out[82]=
$$\frac{5}{2} + x^2 - 2\, y + y^2 + z^2 - \frac{1}{2}\sqrt{5 + 16\, x^2 + 16\, x\, (-1 + y) - 8\, y + 4\, y^2 + 4\, z^2}$$

In[83]:=
```
DzsMin = Together [D[sMin , z]]
```

Out[83]=
$$\frac{2\, z\left(-1 + \sqrt{5 - 16\, x + 16\, x^2 - 8\, y + 16\, x\, y + 4\, y^2 + 4\, z^2}\right)}{\sqrt{5 - 16\, x + 16\, x^2 - 8\, y + 16\, x\, y + 4\, y^2 + 4\, z^2}}$$

In[84]:=
```
Simplify [Expand [(4 x + 2 y - 2) ^ 2 + 4 z ^ 2 + 1]]
```

Out[84]=
$$5 + 16\, x^2 + 16\, x\, (-1 + y) - 8\, y + 4\, y^2 + 4\, z^2$$

In[85]:=
```
Q = ReplaceAll [Q, z → 0]
```

Out[85]=
$$\{\{3 + x^2 + (-2 + y)\, y,\ 1 - 2\, x - y\},\ \{1 - 2\, x - y,\ 2 + x^2 + (-2 + y)\, y\}\}$$

In[86]:=
```
{evalQ , evecQ} = Simplify [Eigensystem [Q], x ∈ Reals && y ∈ Reals && z ∈ Reals]
```

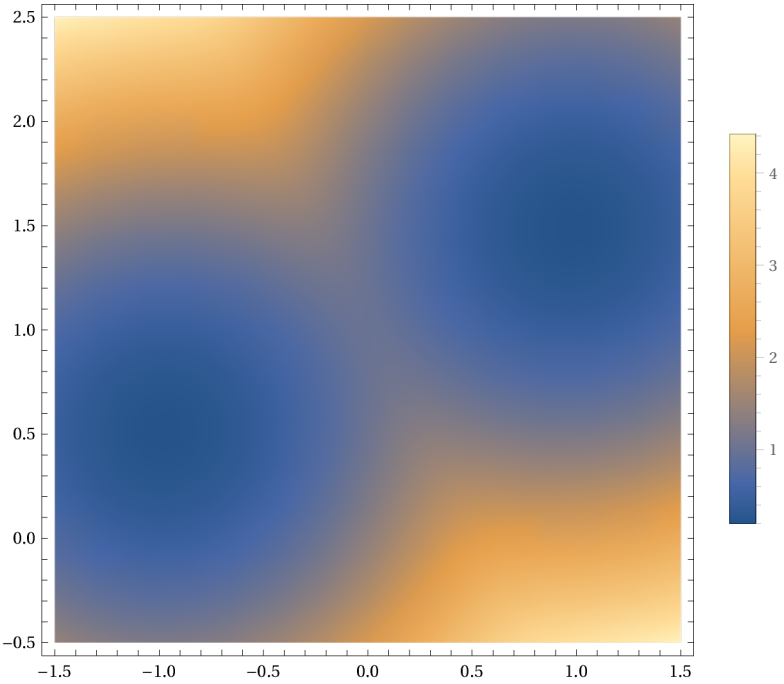
Out[86]=
$$\left\{\left\{\frac{5}{2} + x^2 - 2\, y + y^2 - \frac{1}{2}\sqrt{5 + 16\, x^2 + 16\, x\, (-1 + y) - 8\, y + 4\, y^2},\right.\right.$$
$$\left.\frac{5}{2} + x^2 - 2\, y + y^2 + \frac{1}{2}\sqrt{5 + 16\, x^2 + 16\, x\, (-1 + y) - 8\, y + 4\, y^2}\right\},$$
$$\left.\left\{\left\{\frac{-1 + \sqrt{5 + 16\, x^2 + 16\, x\, (-1 + y) - 8\, y + 4\, y^2}}{2 \times (-1 + 2\, x + y)},\ 1\right\},\ \left\{-\frac{1 + \sqrt{5 + 16\, x^2 + 16\, x\, (-1 + y) - 8\, y + 4\, y^2}}{2 \times (-1 + 2\, x + y)},\ 1\right\}\right\}\right\}$$

In[87]:=
```
sMin = ReplaceAll[sMin, z → 0]
```

Out[87]=
$$\frac{5}{2} + x^2 - 2\,y + y^2 - \frac{1}{2}\sqrt{5 + 16\,x^2 + 16\,x\,(-1 + y) - 8\,y + 4\,y^2}$$

In[88]:=
```
DensityPlot[sMin, {x, -1.5, 1.5}, {y, -0.5, 2.5}, PlotLegends → True]
```

Out[88]=




In[89]:=
```
minOnGraph = Minimize[{sMin, x > 0}, {x, y}]
```

Out[89]=
$$\left\{\frac{1}{5}, \left\{x \to \frac{2\sqrt{6}}{5}, y \to \frac{1}{5} \times \left(5 + \sqrt{6}\right)\right\}\right\}$$

In[90]:=
```
minOnGraph = Minimize[{sMin, x < 0}, {x, y}]
```

Out[90]=
$$\left\{\frac{1}{5}, \left\{x \to -\frac{2\sqrt{6}}{5}, y \to \frac{1}{5} \times \left(5 - \sqrt{6}\right)\right\}\right\}$$

In[91]:=
```
Qmin = ReplaceAll[Q, minOnGraph[[2]]];
```

In[92]:=
```
MatrixForm[Qmin]
```

Out[92]//MatrixForm=
$$\begin{pmatrix} \frac{99}{25} + \frac{1}{5} \times \left(5 - \sqrt{6}\right) \times \left(-2 + \frac{1}{5} \times \left(5 - \sqrt{6}\right)\right) & 1 + \frac{4\sqrt{6}}{5} + \frac{1}{5} \times \left(-5 + \sqrt{6}\right) \\ 1 + \frac{4\sqrt{6}}{5} + \frac{1}{5} \times \left(-5 + \sqrt{6}\right) & \frac{74}{25} + \frac{1}{5} \times \left(5 - \sqrt{6}\right) \times \left(-2 + \frac{1}{5} \times \left(5 - \sqrt{6}\right)\right) \end{pmatrix}$$

In[93]:=
```
{sMin, vec} = Eigensystem[Qmin]
```

Out[93]=
$$\left\{\left\{\frac{26}{5}, \frac{1}{5}\right\}, \left\{\left\{\sqrt{\frac{3}{2}}, 1\right\}, \left\{-\sqrt{\frac{2}{3}}, 1\right\}\right\}\right\}$$

In[94]:=
```
sMin = sMin[[2]]
```

Out[94]=
$$\frac{1}{5}$$

In[95]:=
```
vec = vec[[2]]
```

Out[95]=
$$\left\{-\sqrt{\frac{2}{3}}, 1\right\}$$

In[96]:=
```
N[sMin]
```

Out[96]=
```
0.2
```

In[97]:=
```
Simplify[(1 / Norm[vec]) vec]
```

Out[97]=
$$\left\{-\sqrt{\frac{2}{5}}, \sqrt{\frac{3}{5}}\right\}$$